

stpois: Fast Poisson event-history regression with high-dimensional fixed effects

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Abstract Poisson regression on episode-split survival data is a workhorse of event-history analysis, but two practical obstacles limit its use with full-population register data: models with rich fixed-effect structures become slow or infeasible to estimate with explicit dummy variables, and continuous covariates prevent the collapsing of individual records to risk cells that otherwise makes such models fast. This article introduces the community-contributed command `stpois`, which fits Poisson event-history models directly on `stset` data through three estimation paths, all of which return the exact maximum-likelihood estimates: a standard path that wraps `poisson` with automatic exposure; a high-dimensional fixed-effects path implemented via iteratively reweighted alternating projections in Mata; and a fast path that accelerates estimation through the cell structure of the categorical covariates, computing each Newton iteration from exponentially tilted within-cell moments in a single pass over the microdata. Coefficients, log likelihoods, and OIM, robust, and cluster-robust standard errors are numerically identical to the corresponding `poisson` model on every path, and factor-variable interactions are supported throughout. Benchmarks against `streg` and `poisson` document the speed gains on episode-split workflows, on all-categorical rate models, and on models with high-dimensional fixed effects.

1 Introduction

The equivalence between piecewise-constant hazard models and Poisson regression on episode-split data is one of the older and more useful results in event-history analysis (Holford 1980; Laird and Olivier 1981; Whitehead 1980). Splitting each subject’s time at risk into episodes, treating the event indicator in each episode as a Poisson count, and entering the log of exposure time as an offset yields maximum-likelihood estimates of a proportional-hazards model with a hazard that is constant within episodes. In Stata, this model can be fitted with `poisson` after manually generating the exposure variable, or — for the single-episode exponential case — with `streg, distribution(exponential)`. The approach is standard in epidemiology (Clayton and Hills 1993; Carstensen 2007) and is covered at length by Rabe-Hesketh and Skrondal (2022).

Two practical obstacles arise when this framework meets modern administrative data. Full-population registers of the kind available in the Scandinavian countries routinely produce analysis files with millions or tens of millions of episode records, particularly after splitting on calendar time, age, and duration. First, models with rich fixed-effect structures — birth cohorts, municipalities, detailed occupational codes, or their interactions — become slow or infeasible when the fixed effects are entered as explicit dummy variables, since estimation time and memory grow with the square of the parameter count. The

`ppmlhdfc` command (Correia et al. 2020) solves this problem for general Poisson models, building on the alternating-projections approach of Guimarães and Portugal (2010), but it is an external dependency with its own installation chain, which matters in the locked-down secure environments where register data must typically be analyzed. Second, and less widely appreciated, the Poisson likelihood for purely categorical covariates depends on the data only through the sums of events and exposure within covariate cells. Estimation on the collapsed cell data is then *exact* and often several orders of magnitude faster. A single continuous covariate breaks this property, and analysts face an unattractive choice between discretizing the covariate and giving up the speed advantage entirely.

The command described in this article, `stpois`, addresses both obstacles within a single interface for `stset` data. It offers three estimation paths, all of which return the exact maximum-likelihood estimates with exact standard errors:

1. A **standard path** that wraps `poisson`, using the event indicator `_d` as outcome and time at risk `_t - _t0` as exposure. Coefficients and standard errors are identical to `streg, distribution(exponential)`.
2. A **high-dimensional fixed-effects path**, `absorb()`, implementing iteratively reweighted least squares with alternating-projections demeaning in Mata — the same mathematical approach as `ppmlhdfc`, but self-contained with no dependencies. Absorbed terms may be single variables or interactions such as `absorb(region#period)`. OIM, robust, and cluster-robust standard errors are numerically identical to those from the corresponding dummy-variable model.
3. A **fast path**, option `fast`, which accelerates exact estimation through the cell structure of the categorical covariates. Newton iterations on the full likelihood are computed from exponentially tilted within-cell moments of the individual-level covariates, accumulated in a single pass over the microdata per iteration; all remaining algebra runs on the cells. When every covariate is categorical, the iterations run entirely on the collapsed cell sums.

Factor-variable interactions — `continuous#continuous`, `continuous#categorical`, and `categorical#categorical` — are supported on all paths. Section 2 sets out the modeling framework, section 3 describes the fast path, and section 4 describes the fixed-effects algorithm. Sections 5 and 6 document the syntax and illustrate the command, and section 7 verifies the numerical claims and benchmarks the command against `streg` and `poisson`.

2 Poisson event-history regression

Consider `stset` data in which record i contributes an event indicator $d_i \in \{0, 1\}$ (Stata's `_d`) and a time at risk $t_i = _t - _t0$. Under a proportional-hazards model with hazard rate $\lambda_i = \exp(\mathbf{x}'_i \boldsymbol{\beta})$ constant within records, the log likelihood for right-censored data is, up to a constant,

$$\ell(\boldsymbol{\beta}) = \sum_i \{d_i \mathbf{x}'_i \boldsymbol{\beta} + d_i \log t_i - t_i \exp(\mathbf{x}'_i \boldsymbol{\beta})\},$$

which is proportional to the likelihood of a Poisson model for d_i with mean $\mu_i = t_i \exp(\mathbf{x}'_i \boldsymbol{\beta})$, that is, with $\log t_i$ as offset. Hence any software that fits Poisson regression with an offset fits this hazard model, and duration dependence or period effects are accommodated by episode splitting (`stsplit`) followed by indicator variables for the resulting time bands (Holford 1980; Laird and Olivier 1981). The Poisson “counts” here are 0/1 event indicators; nothing requires the outcome to be a genuine count, and the pseudo-maximum-likelihood results of Gourieroux et al. (1984) imply that the coefficient estimates are consistent whenever the log-linear mean is correctly specified, with robust standard errors available in the usual way (see also Santos Silva and Tenreiro 2006).

`stpois` fits exactly this model on the current `stset` data. On the standard path it constructs the exposure variable, calls `poisson`, and posts results with survival-flavored headers (numbers of subjects and failures, time at risk). The point of the command, however, is not the convenience wrapper but the two departures from it, to which I now turn.

2.1 Sufficiency of cell sums under categorical covariates

When all covariates are categorical, the covariate vector takes a finite number of distinct values, indexing cells $j = 1, \dots, J$. The log likelihood above then depends on the data only through the within-cell sums $D_j = \sum_{i \in j} d_i$ and $T_j = \sum_{i \in j} t_i$:

$$\ell(\beta) = \sum_j \{D_j \mathbf{x}'_j \beta - T_j \exp(\mathbf{x}'_j \beta)\} + \text{const}.$$

(D_j, T_j) are sufficient statistics, and estimation on the J collapsed cells reproduces the individual-level maximum-likelihood estimates exactly — coefficients, standard errors, and all. With millions of episode records but a few hundred cells (period $\times$ education $\times$ region, say), the gain is dramatic, and this is the standard trick behind register-based rate modeling (Clayton and Hills 1993). A continuous covariate destroys the sufficiency because $\exp(\gamma x_i)$ no longer aggregates: $\sum_{i \in j} t_i \exp(\gamma x_i) \neq T_j \exp(\gamma \bar{x}_j)$ by Jensen's inequality. The fast path generalizes the collapsing idea so that it delivers the exact MLE in the presence of continuous covariates as well.

3 Fast exact estimation: the fast option

3.1 Term classification and cells

With `fast`, `stpois` classifies each expanded term of the varlist by its pieces. Terms built entirely from factor-variable pieces — `i.edu`, `i.edu#i.region` — are constant within cells and enter the model at the cell level, with coefficient vector δ on the cell-level design $\mathbf{w}_j$. Terms involving at least one continuous piece — `age`, `c.age#c.income`, `c.age#i.edu` — enter at the individual level, with coefficient vector γ on the individual-level design $\mathbf{X}_i$, so that $\mu_i = t_i \exp(\gamma' \mathbf{X}_i + \delta' \mathbf{w}_{j(i)})$. The cells j are the cross-classification of the categorical variables appearing in the cell-level terms; at least one cell-level term is required.

3.2 Newton iterations on tilted cell moments

The score and observed information of the Poisson likelihood decompose into pieces that are either fixed cell sums or *exponentially tilted within-cell moments* of the individual-level covariates:

$$M_j = \sum_{i \in j} \mu_i, \quad \mathbf{m}_j = \sum_{i \in j} \mu_i \mathbf{X}_i, \quad \mathbf{Q} = \sum_i \mu_i \mathbf{X}_i \mathbf{X}_i'.$$

The score is $\left(\sum_i d_i \mathbf{X}_i - \sum_j \mathbf{m}_j, \sum_j \mathbf{w}_j (D_j - M_j) \right)$ and the information matrix has blocks $\mathbf{Q}$, $\sum_j \mathbf{m}_j \mathbf{w}_j'$, and $\sum_j M_j \mathbf{w}_j \mathbf{w}_j'$. Each Newton iteration therefore requires one pass over the microdata to accumulate the tilted moments — at cost $O(np^2)$, with p the number of *individual-level* columns only — after which everything happens on the J cells. A full-model Newton step in `poisson`, by contrast, costs $O(n(p+k)^2)$ with k the number of cell-level parameters, so the per-iteration floating-point saving is a factor of roughly $\{(p+k)/p\}^2$, and it grows with the richness of the categorical structure. Step-halving on the log likelihood guards the updates, and convergence is declared when $\max |\Delta \theta| / (1 + |\theta|)$ falls below `tolerance()`.

When *every* term is cell-level ($p = 0$), the sufficiency result of section 2.1 applies directly, and the iterations run entirely on the collapsed sums (D_j, T_j) : after a single aggregation pass, each Newton step costs $O(Jk^2)$ regardless of the number of underlying episode records.

Because this is Newton on the exact likelihood, nothing is approximated: coefficients, the log likelihood, and the inverse-information (OIM) standard errors are numerically identical to the full `poisson` model, and robust and cluster-robust variance estimators are computed from the same per-observation scores as their `poisson` counterparts, with the HC1 and $G/(G - 1)$ small-sample corrections. Because the continuous coefficients are estimated from the full individual-level likelihood, they are identified — with full efficiency — even when the continuous covariates do not vary across cells at all.

3.3 Implementation

The data handling is designed for large datasets, adopting the strategy popularized by the `ftools` project (Correia 2016): group structures are computed from a single `order()` sort in Mata, segment boundaries are found by comparing adjacent sorted rows, and all within-group sums — the cell aggregates, the tilted moments, and the within-cluster score sums of the cluster-robust variance estimator — are running-sum differences over the sorted data, each an $O(n)$ pass independent of the number of groups. Interaction columns are constructed directly from the raw variables in Mata. There are no `preserve/collapse` round-trips and no temporary variables in the dataset; beyond the single sort permutation, memory overhead is one copy of the loaded columns.

4 High-dimensional fixed effects: `absorb()`

The `absorb(terms)` option fits the exact model with one or more sets of fixed effects without creating dummy variables, using iteratively reweighted least squares (IRLS) combined with weighted alternating-projections demeaning — the approach introduced to Stata by Guimarães and Portugal (2010) and refined by Correia et al. (2020). Each absorbed term is a variable or an interaction such as `region#period`, absorbed as the cross-classification of its variables. Each IRLS iteration:

1. forms the working variable $z_i = \eta_i + (d_i - \mu_i)/\mu_i$ and weights $w_i = \mu_i$, where η_i is the current linear predictor including the offset and $\mu_i = t_i \exp(\eta_i)$;
2. demeans z and each covariate within the groups of each absorbed term in turn, iterating this Gauss-Seidel cycle to convergence (weighted alternating projections);
3. regresses the demeaned $\tilde{z}$ on the demeaned $\tilde{\mathbf{X}}$ by weighted least squares to update β ; and
4. recovers the fixed-effect contribution as the complement of the projection and updates η .

Convergence is declared when $\max_k |\Delta\beta_k| / (1 + |\beta_k|)$ falls below `tolerance()` (default 10^{-8}). Sort permutations and group boundaries for each absorbed term are precomputed once, so every demeaning pass is a single $O(n)$ sweep independent of the number of levels. The implementation is pure Mata with no dependencies, and the non-absorbed varlist may include factor variables and interactions.

Standard errors come from the partitioned information at convergence: $V = (\tilde{\mathbf{X}}'W\tilde{\mathbf{X}})^{-1}$ for `vce(oim)`, the HC1 sandwich with meat $\sum_i s_i s_i' \cdot n/(n - 1)$ and scores $s_i = (d_i - \mu_i)\tilde{x}_i$ for `vce(robust)`, and the $G/(G - 1)$ -corrected clustered sandwich with scores summed within clusters for `vce(cluster clustvar)`. By the Frisch-Waugh-Lovell logic these are not approximations: all three match the corresponding dummy-variable `poisson` output at numerical precision, which the test suite verifies (section 7).

Groups of an absorbed term in which no events occur are separated in the maximum-likelihood sense — their fixed effects diverge to $-\infty$ — and `stpois` drops such observations before estimation with a message, mirroring the practice of `ppmlhdfc` (Correia et al. 2020, who treat separation in far greater generality).

5 The stpois command

5.1 Syntax

```
stpois varlist [if] [in] [weight] [, irr nolog level(#) noconstant  
      vce(vcetype) robust cluster(varname)  
      absorb(terms) tolerance(#) maxiter(%)  
      fast poisson_options ]
```

The data must be `stset`; see [ST] `stset`. `fweights`, `iwweights`, and `pweights` are allowed. `fast` and `absorb()` may not be combined. The varlist may include factor variables and interactions on every path.

5.2 Options

irr reports exponentiated coefficients (incidence-rate ratios, equal to hazard ratios in this model). Display only.

nolog suppresses iteration logs.

level(#) sets the confidence level; the default is `level(95)`.

noconstant suppresses the constant term.

vce(vcetype), **robust**, and **cluster(varname)** specify the variance estimator. `vce(oim)` (the default), `vce(robust)`, and `vce(cluster clustvar)` are supported on all paths and are numerically exact relative to the corresponding full `poisson` model; the standard path additionally accepts anything `poisson` accepts.

fast selects cell-accelerated exact estimation; see section 3. With `fast`, terms built only from factor-variable pieces define the cells and enter at the cell level; at least one such term is required.

absorb(terms) absorbs the fixed effects of one or more terms, each a variable name or a `var1#var2` interaction; see section 4.

tolerance(#) and **maxiter(#)** control the iteration convergence criterion and cap for `absorb()` and `fast`; defaults are `1e-8` and `100`.

Any remaining options on the standard path are passed to `poisson`.

5.3 Stored results

`stpois` stores the usual `e(b)`, `e(V)`, `e(sample)`, log likelihood, and model-test scalars, plus survival-specific scalars `e(N_sub)` (subjects), `e(N_fail)` (failures), and `e(risk)` (total time at risk). The fast path adds `e(N_cells)` and the macros `e(cont_vars)` and `e(cat_vars)`; `absorb()` adds `e(absorb)` and, with clustering, `e(N_clust)` and `e(clustvar)`. All paths report the individual-level log likelihood, so likelihood-based statistics are comparable across paths. `e(predict)` is set to `stpois_p`, so `predict` offers the linear predictor (`xb`, the default, including the log-exposure offset unless `nooffset`), predicted numbers of events (`n`), hazard rates (`hazard` or `hr`), and survival probabilities (`survival`). Standard postestimation commands (`margins`, `lincom`, `test`, `estimates`) work as usual.

6 Examples

6.1 Standard path

The default path reproduces `streg`, `distribution(exponential)` exactly (the log likelihoods differ by a constant because `streg` includes the $d_i \log t_i$ terms):

```
. webuse stan3
. stset t1, failure(died) id(id)
. stpois age posttran, nolog
```

Poisson event-history regression

No. of subjects =	103	Number of obs =	172
No. of failures =	75		
Time at risk =	31938	LR chi2(2) =	36.75
Log likelihood =	-233.83404	Prob > chi2 =	0.0000

	_d	Coefficient	Std. err.	z	P> z	[95% conf. interval]
	+					
	age	.0605817	.015235	3.98	0.000	.0307216 .0904417
	posttran	-1.285206	.2377048	-5.41	0.000	-1.751099 -.8193135
	_cons	-7.8376	.7094787	-11.05	0.000	-9.228153 -6.447047

6.2 Absorbing fixed effects

`absorb()` returns the same coefficients and standard errors as the explicit dummy-variable model, without estimating the absorbed effects:

```
. stpois age posttran, absorb(surgery) nolog
```

Poisson EHA – HD FE (absorb: surgery)

No. of subjects =	103	Number of obs =	172
No. of failures =	75		
Time at risk =	31938	LR chi2(2) =	49.18
Log likelihood =	-227.61949	Prob > chi2 =	0.0000

	_d	Coefficient	Std. err.	z	P> z	[95% conf. interval]
	+					
	age	.0610952	.014454	4.23	0.000	.0327659 .0894244
	posttran	-1.141144	.2402034	-4.75	0.000	-1.611934 -.6703538

For comparison, `poisson _d age posttran i.surgery, exposure(exp)` on the same sample gives age = .0610952 (SE .014454) and posttran = -1.141144 (SE .2402034): identical to all printed digits.

6.3 Fast estimation on a large dataset

The following simulation generates 500,000 subjects in 200 risk cells (4 education × 5 region × 10 period), with two continuous covariates (age, income):

```
. clear
. set obs 500000
. set seed 20260713
```

```

. gen edu    = ceil(4 * runiform())
. gen region = ceil(5 * runiform())
. gen period = ceil(10 * runiform())
. gen age    = 30 + 20 * runiform()
. gen income = rnormal(10, 2)
. gen lambda = exp(-6 + 0.03*age + 0.05*income
>   + 0.3*(edu==2) + 0.5*(edu==3) - 0.2*(edu==4)
>   + 0.1*(region==2) - 0.2*(region==3) + 0.2*(period >= 6))
. gen ptime = 0.5 + 0.5 * runiform()
. gen failed = (runiform() < 1 - exp(-lambda * ptime))
. gen id2    = _n
. stset ptime, failure(failed) id(id2)

```

```

. stpois age income i.edu i.region i.period, fast nolog

```

Poisson EHA – fast (cell-accelerated exact MLE)

No. of subjects =	500,000	Number of obs =	500,000
No. of failures =	6,759		
Time at risk =	375045	LR chi2(18) =	850.69
Log likelihood =	-35297.23308	Prob > chi2 =	0.0000

```

-----+-----
      _d | Coefficient   Std. err.      z    P>|z|    [95% conf. interval]
-----+-----
      age |   .0265379    .002121   12.51   0.000    .0223808    .0306951
    income |   .0518263    .0060872    8.51   0.000    .0398956    .0637571
          |
      edu |
        2 |   .3070999    .0353078    8.70   0.000    .237898    .3763019
        3 |   .5094171    .0339083   15.02   0.000    .442958    .5758762
        4 |  -.1907793    .0398372   -4.79   0.000   -.2688587   -.1126998
(output omitted)
-----+-----

```

Running the same model on the standard path returns identical output — the same coefficients, standard errors, and log likelihood to all digits — in five Newton iterations versus a full `poisson` maximization; the timing comparison is in section 7.

6.4 Interactions

All interaction types are accepted with `fast`; terms involving a continuous piece, such as `edu#c.age` below, are estimated at the individual level, while the categorical main effects and any categorical#category interactions run at the cell level:

```

. stpois c.age##i.edu income i.region i.period, fast nolog

```

(output omitted)

```

-----+-----
      age |   .0255056    .0046717    5.46   0.000    .0163492    .0346621
          |

```



```

edu#c.age |
      2 | -.0009954   .0061531   -0.16   0.871   -.0130552   .0110645
      3 |  .0021302   .0059106    0.36   0.719   -.0094545   .0137148
      4 |  .0034035   .0069472    0.49   0.624   -.0102128   .0170198
(output omitted)

```

Interacted fixed effects are absorbed with `absorb(edu#region)`, which matches `poisson` with explicit `i.edu#i.region` dummies on coefficients and standard errors.

6.5 Episode-split workflow

In applications, the typical sequence splits spells on calendar time, age, or duration and then estimates piecewise-constant hazard models:

```

. stsplot fu, at(1(1)30)
. stpois i.fu age income i.educ, irr
. stpois i.fu age income i.educ, fast
. stpois age income, absorb(fu educ region) vce(cluster region)

```

7 Validation and performance

7.1 Numerical agreement with `streg` and `poisson`

The package ships with a test suite (`tests/test_stpois.do` and `tests/test_fast_exact.do`) that runs on the `stan3` heart-transplant data and on simulated data. Each exactness claim is held to a numerical benchmark:

1. The **standard path** matches `streg`, `distribution(exponential)` on every coefficient and standard error to 10^{-6} . On the `stan3` data, both report $\hat{\beta}_{\text{age}} = 0.0605817$ (SE 0.0152350) and $\hat{\beta}_{\text{posttran}} = -1.2852060$ (SE 0.2377048); the log likelihoods differ only by the fixed $\sum_i d_i \log t_i$ term that `streg` includes and the Poisson formulation drops.
2. **absorb()** matches the explicit dummy-variable `poisson` model on coefficients *and* standard errors (observed agreement on the order of 10^{-8}), for OIM and robust variance estimators, with the cluster estimator covered by the same sandwich code path. The check covers single fixed effects, interacted fixed effects (`absorb(edu#region)` against `i.educ#i.region` dummies), and factor-variable interactions in the varlist.
3. **fast** matches the full `poisson` model on coefficients, standard errors (OIM, robust, and clustered), and log likelihood to 10^{-6} . The check covers continuous#continuous, continuous#categorical, and categorical#categorical interactions, all-categorical models, and a two-cell setting in which the continuous coefficient is identified purely from within-cell variation.

All tests pass on Stata 18/MP; the command requires version 14.

7.2 Timing benchmarks

All timings are single runs on Stata MP (Apple silicon), produced by the scripts in the package repository; in every comparison the estimates agree across methods to the tolerances above, so the tables compare identical output.

Table 1 times a mixed model with two continuous covariates and 18 categorical parameters (education, region, period), comparing `streg`, `distribution(exponential)`, the `stpois` standard path (a `poisson` call), and `stpois, fast`:

n	streg	stpois (standard)	fast	speedup vs. streg
100,000	0.42 s	0.35 s	0.16 s	2.6×
1,000,000	2.98 s	2.67 s	1.21 s	2.5×
5,000,000	14.23 s	12.79 s	7.38 s	1.9×

Table 2 times the canonical episode-split workflow: 100,000 subjects observed for up to 30 years are split into one-year episodes with `stsplit` (1,888,654 episode records), and a piecewise-constant hazard model with 30 follow-up bands, education, region, and two continuous covariates is estimated on the episode file:

Method	Time
streg, distribution(exponential)	5.72 s
stpois (standard)	4.97 s
stpois, fast	2.06 s

Table 3 times an all-categorical rate model (education × region × period, 200 cells), the case where the sufficiency result of section 2.1 lets `fast` iterate on the collapsed cell sums alone:

n	streg	stpois (standard)	fast
1,000,000	2.61 s	2.42 s	0.78 s
5,000,000	12.82 s	11.80 s	5.41 s

In tables 1–3 the remaining `fast` cost is dominated by loading and sorting the microdata once; the Newton iterations themselves are essentially free. The gains are therefore largest relative to estimators that must pass over the microdata repeatedly, which is what the fixed-effects benchmark isolates. Table 4 estimates a model with two continuous covariates and a single fixed effect with 1,000 levels on one million observations, comparing `poisson` with explicit dummies against `stpois`, `absorb()`:

Method	Time	$\hat{\beta}_{\text{age}}$	SE
<code>poisson</code> with 1,000 dummies	32.9 s	0.0266067	0.0017000
<code>stpois</code> , <code>absorb(g)</code>	3.2 s	0.0266067	0.0017000

A tenfold saving at identical output; the dummy-variable model’s cost grows roughly quadratically in the number of levels while the absorption cost grows only linearly, so the gap widens further at higher dimensions. For register applications with 10^7 or more episodes, detailed period × education × region structures, and high-dimensional nuisance effects, these two mechanisms — cell acceleration and fixed-effect absorption — turn minutes into seconds, which changes what is practical to do interactively: specification searches, bootstrap loops, or estimation inside secure environments with tight session limits.

8 Conclusion

`stpois` packages three things that belong together but have required separate workflows: the Poisson formulation of event-history models on `stset` data, absorption of high-dimensional fixed effects without external dependencies, and cell-accelerated estimation that retains continuous covariates. Every estimation path is exact in the strict sense — the test suite holds coefficients *and* variance estimators, not

merely point estimates, to the `streg` and `poisson` benchmarks — and factor-variable interactions are supported throughout, including interacted fixed effects. The command, its help file, the test suite, and the benchmark scripts are available from the author’s repository and are installable with `net install`.

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